

Exercise 1.

$$P(C_F) = \frac{9}{10}$$

$$P(C_N) = \frac{1}{10}$$

$$P(H_4|C_F) = \left(\frac{1}{2}\right)^2 = \frac{1}{16}$$

$$P(H_4|C_N) = 1$$

$$P(H_4) = \frac{9}{10} \times \frac{1}{16} + 1 \times \frac{1}{10} = \frac{9}{160} + \frac{1}{10} = \frac{5}{32}$$

The probability the coin is double headed based on 4 heads in a row:

$$P(C_N|H_4) = \frac{P(H_4|C_N)P(C_N)}{P(H_4)} = \frac{1 \times \frac{1}{10}}{\frac{5}{32}} = \frac{\frac{1}{10}}{\frac{5}{32}} = \frac{1}{10} \times \frac{32}{5} = \frac{32}{50} = \frac{16}{25}$$

$$P(C_F|H_4) = \frac{P(H_4|C_F)P(C_F)}{P(H_4)} = \frac{\frac{1}{16} \times \frac{9}{10}}{\frac{5}{32}} = \frac{\frac{9}{160}}{\frac{5}{32}} = \frac{9}{160} \times \frac{32}{5} = \frac{9}{25}$$

$$P(H|H_4) = P(C_F|H_4) \times \frac{1}{2} + P(C_N|H_4) \times 1$$

$$P(H|H_4) = \frac{9}{25} \times \frac{1}{2} + \frac{16}{25} \times 1$$

$$= \frac{9}{50} + \frac{16}{25} = \frac{41}{50}$$

Exercise 2.

BOOLAHUBOO

$$B = 2$$

$$O = 4$$

$$L = 1$$

$$A = 1$$

$$H = 1$$

$$U = 1$$

10 letters

$$|\Omega| = \binom{10}{2,4,1,1,1,1} = \frac{10!}{2!4!1!1!1!1!} = \frac{3,628,800}{2 \times 24 \times 1} = \frac{3,628,800}{48} = 75,600$$

$$P(B) = \binom{10}{1,4,1,1,1,1} = \frac{9!}{1!4!1!1!1!1!} = \frac{362,880}{2 \times 24 \times 1} = \frac{362,880}{48} = 15,120$$

There are $5! = 120$ ways to arrange the non O letters.

Between the 5 letters, there are 6 gaps where 1 O can be placed. 4 out of these gaps can be chosen. So,

$$= \binom{6}{4} = 15$$

Therefore,

$$P(B \cap O_B) = 120 \times 15 = 1,800$$

$$P(O_B|B) = \frac{1,800}{15,120} = \frac{5}{42} \approx 0.1190$$

Exercise 3.**(a)**

- one match has n ways of being placed
- for the n-1 elements there is no matches

$$m_1(n) = r_1(n-1) = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^{n-1} \frac{1}{(n-1)!}$$

$$\text{(b)} \quad p(m_1|p_1) = \frac{P(m_1 \cap p_1)}{P(p_1)} = \frac{P(m_1)}{P(p_1)} = \frac{1 - \frac{1}{1!} + \frac{1}{2!} - \dots + \frac{(-1)^{n-1}}{(n-1)!}}{1 - \frac{1}{1!} + \frac{1}{2!} - \dots + \frac{(-1)^{n-1}}{n!}}$$

(c) conditional prob. : $P(r_2|r_1) = 1 - P(m_1|r_1)$

$$P(r_2|r_1) = 1 - \frac{1 - \frac{1}{1!} + \frac{1}{2!} - \dots + \frac{(-1)^{n-1}}{(n-1)!}}{1 - \frac{1}{1!} + \frac{1}{2!} - \dots + \frac{(-1)^{n-1}}{n!}}$$

Exercise 4.

(1, 1) (1, 2) (1, 3) (1, 4) (1, 5) (1, 6)

(2, 1) (2, 2) (2, 3) (2, 4) (2, 5) (2, 6)

(3, 1) (3, 2) (3, 3) (3, 4) (3, 5) (3, 6)

(4, 1) (4, 2) (4, 3) (4, 4) (4, 5) (4, 6)

(5, 1) (5, 2) (5, 3) (5, 4) (5, 5) (5, 6)

(6, 1) (6, 2) (6, 3) (6, 4) (6, 5) (6, 6)

$$P(C_H) = \frac{1}{2}$$

$$P(C_T) = \frac{1}{2}$$

$$P(D_6|C_H) = \frac{1}{6}$$

$$P(D_6|C_F) = \frac{5}{36}$$

$$P(D_6) = P(D_6|C_H) \times P(C_H) + P(D_6|C_T) \times P(C_T)$$

$$= \frac{1}{6} \times \frac{1}{2} + \frac{5}{36} \times \frac{1}{2}$$

$$= \frac{1}{12} + \frac{5}{72}$$

$$= \frac{11}{72}$$

Exercise 5.

6 tennis balls: 4 brand new, 2 used

After someone uses a new ball it becomes a used ball.

(a) Ways the roommate can grab new balls:

- Probability I grabbed 2 new ones and roommate grabbed 2 new ones:

$$\text{Probability 2 new ones grabbed} = P(M_n) = \frac{4}{6} \times \frac{3}{5} = \frac{2}{5}$$

$$\text{Probability roommate grabs 2 new balls} = P(M_n|R_n) = \frac{2}{6} \times \frac{1}{5} = \frac{1}{15}$$

- Probability I grabbed an old and a new one and roommate grabbed 2 new ones:

$$\text{Probability 1 new and 1 old are grabbed} = P(M_{n,o}) = \frac{2}{6} \times \frac{4}{5} = \frac{4}{15}$$

$$\text{Probability roommate grabs 2 new balls} = P(M_{n,o}|R_n) = \frac{3}{6} \times \frac{2}{5} = \frac{1}{5}$$

- Probability I grabbed a new and an old one and roommate grabbed 2 new ones

$$\text{Probability 1 old and 1 new are grabbed} = P(M_{o,n}) = \frac{4}{6} \times \frac{2}{5} = \frac{4}{15}$$

$$\text{Probability roommate grabs 2 new balls} = P(M_{o,n}|R_n) = \frac{3}{6} \times \frac{2}{5} = \frac{1}{5}$$

- Probability I grabbed 2 old ones and roommate grabbed 2 new ones:

$$\text{Probability 2 old ones grabbed} = P(M_o) = \frac{2}{6} \times \frac{1}{5} = \frac{1}{15}$$

$$\text{Probability roommate grabs 2 new balls} = P(M_o|R_n) = \frac{4}{6} \times \frac{3}{5} = \frac{2}{5}$$

- $P(R_n) = P(M_n|R_n)P(M_n) + P(M_{n,o}|R_n)P(M_{n,o}) + P(M_{o,n}|R_n)P(M_{o,n}) + P(M_o|R_n)P(M_o)$

$$P(R_n) = \frac{2}{5} \times \frac{1}{15} + \frac{4}{15} \times \frac{1}{5} + \frac{4}{15} \times \frac{1}{5} + \frac{1}{15} \times \frac{2}{5}$$

$$P(R_n) = \frac{2}{75} + \frac{4}{75} + \frac{4}{75} + \frac{2}{75}$$

$$P(R_n) = \frac{4}{25} = 0.16$$

(b) Roommate grabbed 2 new balls, what is the prob. I used 2 new ones?

$$P(M_n|R_n) = \frac{(R_n|M_n)P(M_n)}{P(R_n)} = \frac{\frac{1}{15} \times \frac{2}{5}}{\frac{4}{25}} = \frac{1}{6}$$

Exercise 6.

$$P(A|B) = \frac{1}{2}$$

$$P(B|A) = \frac{1}{4}$$

$$P(A \cup B) = \frac{1}{4}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{1}{2}$$

$$2P(A \cap B) = P(B)$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{1}{4}$$

$$4P(B \cap A) = P(A) = 4P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{3}{4} = P(A) + P(B) - P(A \cap B)$$

$$\frac{3}{4} = 4P(A \cap B) + 2P(A \cap B) - P(A \cap B)$$

$$\frac{3}{4} = 5P(A \cap B)$$

$$\frac{3}{20} = P(A \cap B)$$

$$P(B) = 2P(A \cap B)$$

$$P(B) = 2 \times \frac{3}{20} = \frac{3}{10}$$

$$P(A) = 4P(A \cap B)$$

$$P(A) = 4 \times \frac{3}{20} = \frac{3}{5}$$

Exercise 7.

(1, 1) (1, 2) (1, 3) (1, 4) (1, 5) (1, 6)

(2, 1) (2, 2) (2, 3) (2, 4) (2, 5) (2, 6)

(3, 1) (3, 2) (3, 3) (3, 4) (3, 5) (3, 6)

(4, 1) (4, 2) (4, 3) (4, 4) (4, 5) (4, 6)

(5, 1) (5, 2) (5, 3) (5, 4) (5, 5) (5, 6)

(6, 1) (6, 2) (6, 3) (6, 4) (6, 5) (6, 6)

8:

(2,6) (3,5) (4,4) (5, 3) (2, 6)

$$P(R_8) = \frac{5}{36}$$

7:

(1, 6) (2, 5) (3, 4) (4, 3) (5, 2) (6, 1)

$$P(R_7) = \frac{6}{36} = \frac{1}{6}$$

No 7 or 8 in a roll:

$$36 - 6 - 5 = 25$$

$$P(R_n) = \frac{25}{36}$$

$$P(R_8 \cap R_7) = 0$$

$$P(R_8 \cap R_7) = \frac{11}{36}$$

Probability the sum of 8 happens before the sum of 7:

$$p = \frac{P(R_8)}{P(R_8 \cap R_7)} = \frac{\frac{5}{36}}{\frac{11}{36}} = \frac{5}{11}$$

Exercise 8.

- Fred hits a bullseye, $P(F) = 0.5$
- Greg hits a bullseye, $P(G) = 0.6$
- Fred and Greg hitting the bullseye is independent

(a) They both hit the bullseye on a throw:

$$P(F \cap G) = P(F)P(G) = 0.5 \times 0.6 = 0.3$$

(b) Exactly one of the hits the bullseye on a throw:

$$P(F \cap G^c) = 0.5 \times 0.4 = 0.2$$

$$P(F^c \cap G) = 0.5 \times 0.6 = 0.3$$

$$P(\text{exactly one hit}) = P(F \cap G^c) + P(F^c \cap G)$$

$$= 0.2 + 0.3 = 0.5$$

(c) Neither hits the bullseye on a throw:

$$P(F^c \cap G^c) = 0.5 \cap 0.4 = 0.2$$

(d) M and A are independent

$$P(M) = P(F_1^c \cap F_2^c \cap F_3^c) = P(F_1^c)P(F_2^c)P(F_3^c) = 0.5 \times 0.5 \times 0.5 = 0.125$$

$$P(A) = P(G_1 \cup G_2 \cup G_3) = P(G_1) + P(G_2) + P(G_3) - P(G_1 \cap G_2) - P(G_1 \cap G_3) - P(G_2 \cap G_3) + P(G_1 \cap G_2 \cap G_3)$$

$$= P(G_1) + P(G_2) + P(G_3) - P(G_1)P(G_2) - P(G_1)P(G_3) - P(G_2)P(G_3) + P(G_1)P(G_2)P(G_3)$$

$$= 0.6 + 0.6 + 0.6 - 0.6 * 0.6 - 0.6 * 0.6 - 0.6 * 0.6 + 0.6 * 0.6 * 0.6$$

$$= 1.8 - 0.36 - 0.36 - 0.36 + 0.216$$

$$= 0.936$$

$$P(M \cap A) = P(M)P(A) = 0.125 \times 0.936 = 0.117$$

Exercise 9.

(a) At least one of them hits:

$$P(H) = 1 - P(F^c \cap G^c)$$

$$P(H) = 1 - P(F^c)P(G^c)$$

$$P(H) = 1 - (0.5)(0.4)$$

$$P(H) = 1 - 0.2$$

$$P(H) = 0.8$$

$$P((F \cap G)|H) = \frac{P((F \cap G) \cap H)}{P(H)} = \frac{P(F \cap H)P(G \cap H)}{P(H)} = \frac{0.5 \times 0.6}{0.8} = \frac{0.3}{0.8} = \frac{3}{8} = 0.375$$

$$(b) P((F^c \cap G)|H) = \frac{P((F^c \cap G) \cap H)}{P(H)} = \frac{P(F^c \cap H)P(G \cap H)}{P(H)} = \frac{0.5 \times 0.6}{0.8} = \frac{0.3}{0.8} = \frac{3}{8} = 0.375$$

$$(c) P(\text{exactly one hit} | H) = \frac{P(1hit)}{P(H)} = \frac{0.5}{0.8} = \frac{5}{8} = 0.625$$

Exercise 10.

- 10 balls, 3 black and 7 white

- color drawn 2 more of that color put back with it

Prob. black balls drawn each of the first 3 trials: $\frac{3}{10} \times \frac{5}{12} \times \frac{7}{14} = \frac{105}{1,680} = 0.0625$

Exercise 11.

- Box A: 3 chips labeled 1-3
- Box B: 6 chips labeled 1-6
- Box C: 12 chips labeled 1-12
- S: Amy selects the same chip twice in a row

$$P(A) = P(B) = P(C) = \frac{1}{3}$$

$$P(S|A) = \frac{1}{3}$$

$$P(S|B) = \frac{1}{6}$$

$$P(S|C) = \frac{1}{12}$$

$$P(S) = P(S|A)P(A) + P(S|B)P(B) + P(S|C)P(C)$$

$$P(S) = \frac{1}{3} \times \frac{1}{3} + \frac{1}{6} \times \frac{1}{3} + \frac{1}{12} \times \frac{1}{3}$$

$$P(S) = \frac{1}{9} + \frac{1}{18} + \frac{1}{36}$$

$$P(S) = \frac{7}{36}$$

$$P(B|S) = \frac{P(S|B)P(B)}{P(S)} = \frac{\frac{1}{6} \times \frac{1}{3}}{\frac{7}{36}} = \frac{2}{7}$$

Exercise 12.

$$P(A \cap B|C) = \frac{P(A \cap B \cap C)}{P(C)}$$

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

$$P(A \cap B|C) = P(A)P(B)$$

$$P(A|C) = \frac{P(A \cap C)}{P(C)} = \frac{P(A)P(C)}{P(C)} = P(A)$$

$$P(B|C) = \frac{P(B \cap C)}{P(C)} = \frac{P(B)P(C)}{P(C)} = P(B)$$

$$P(A \cap B|C) = P(A)P(B) = P(A|C)P(B|C)$$

$$P(A \cap B|C) = P(A|C)P(B|C)$$